

MFT (Multiple Choice Questions)

1. A PDA chooses the next move based on [BL-1]
 - a) Current state
 - b) Next input symbol
 - c) Both (a) and (b)
 - d) None of these
2. Which type of symbols contain in the stack of PDA [BL-1]
 - a) Variable
 - b) Terminal
 - c) Both (a) and (b)
 - d) None of these
3. A push down automata can be represented using: [BL-1]
 - a) Transition graph
 - b) Transition table
 - c) ID
 - d) All of the mentioned
4. A language is accepted by a push down automata if it is: [BL-2]
 - a) regular
 - b) context free
 - c) both a and b
 - d) none of the mentioned
5. The following move of PDA is on the basis of: [BL-2]
 - a) Present space
 - b) input symbol
 - c) both a and b
 - d) none of the mentioned

6. A string is accepted by a PDA when... [BL-1]
- a) Stack is empty
 - b) acceptance state
 - c) both a and b
 - d) none of the mentioned
7. Which of the operation are eligible in PDA [BL-1]
- a) push/pop
 - b) delete
 - c) insert
 - d) none of these
8. Push down automata accepts _____ languages [BL-1]
- a) Type 3
 - b) Type 2
 - c) Type 1
 - d) Type 0
9. A push down automaton employs _____ data structure [BL-1]
- a) Queue
 - b) Linked List
 - c) Hash Table
 - d) Stack
10. Which among the following is a CFG for the given Language
 $L = \{x \in \{0,1\}^* \mid \text{number of zeros in } x = \text{number of one's in } x\}$ [BL-2]
- a) $S \rightarrow 0S1 \mid 1S0 \mid SS \mid \epsilon$
 - b) $S \rightarrow 0B \mid 1A \mid A \rightarrow 0S \mid B \rightarrow 1S$
 - c) All of the mentioned
 - d) none of these`
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Long Answer

11. Give leftmost and rightmost derivations of the following strings:
 - a) 00101
 - b) 1001
 - c) 00011 [BL-3]
12. Construct PDA which accepts the language $L = \{a^n c w^n \mid w \in \{a, b\}^*, c \in \Sigma\}$ [BL-3]
13. Construct PDA for the language $L = \{a^n b^n \mid a, b \in \{a, b\}^*, n \geq 0\}$ [BL-3]
14. Design a PDA to accept the set of all strings of 0's and 1's such that no prefix has more 1's than 0's [BL-3]
15. Construct PDA: Accepting the set of all strings over $\{a, b\}$ with equal number of a's and b's. Show the moves for **abbaba** [BL-3]
16. Construct PDA to accept by final state the language of all strings of 0's and 1's such that number of 1's is less than number of 0's. Also convert the PDA to accept by empty stack [BL-3]
17. Enlist the steps to convert PDA to CFG with suitable example [BL-2]
18. Explain push down automata with an example [BL-2]
19. Consider the following CFG: $G = (\{S, A\}, \{a, b\}, P, S)$ Where P consists of:
 $S \rightarrow aAs \mid a$
 $A \rightarrow SbA \mid ss \mid ba$
Derive string 'aabbba' using leftmost & rightmost derivation
20. Convert given CFG into GNF [BL-3]:
 $S \rightarrow Bs \mid Aa$

$$A \rightarrow Bc$$

$$B \rightarrow Ac \text{ where, } V = \{S, A, B\} \text{ \& } T = \{a, c\}$$

21. Eliminate the ϵ -productions from the Grammar G which is defined as:

[BL-3]

$$S \rightarrow ABA$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

22. Design CFG for the following Languages [BL-3]

$$\text{i) } L = \{a^i b^j c^k \mid i = j + k\}$$

$$\text{ii) } L = \{a^{2n} b^n c^n \mid n \geq 1\}$$

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23. Consider CFG with productions [BL-3]

$$S \rightarrow aB/bA$$

$$A \rightarrow a/aS/bAA$$

$$B \rightarrow b/bS/aBB$$

For string “**aaabbabbba**” find leftmost and rightmost derivation.

24. Convert the given regular grammar to its equivalent FA [BL-3]

$$S \rightarrow aS / bS / aA$$

$$A \rightarrow bB$$

$$B \rightarrow aC$$

$$C \rightarrow a$$

25. Define CNF and GNF with suitable examples [BL-2]

MFT-04 (Multiple Choice Questions)

1. Turing machine was invented by: [BL-1]
 - a) Alan Turing
 - b) Turing man
 - c) Turing theory
 - d) Noam Chomsky
2. Turing machine is more powerful than: [BL-1]
 - a) Finite automata
 - b) Push down automata
 - c) Both (a) and (b)
 - d) None of these
3. In one move the Turing machine: [BL-2]
 - a) May change its state
 - b) Write a symbol on the cell being scanned
 - c) Move the head one position left or right
 - d) All of the above
4. A Universal Turing Machine (UTM) can simulate any other Turing Machine (True/False) [BL-1]
5. The Turing Church Thesis states that Turing Machines can solve any computable problem (True/False) [BL-1]
6. Find true or false statement. [BL-2]
 - a) Turing machine was developed by Alan Turing
 - b) PDA is less powerful than Turing machine
 - c) Both (a) and (b)
 - d) None of these

7. The value of n if Turing machine is defined using n -tuples: [BL-1]
- a) 3
 - b) 5
 - c) 6
 - d) $2L+1$
8. If δ is not defined on the current state and the current tape symbol, then the machine ____
- a) Does not halts
 - b) Halts
 - c) goes into loop forever
 - d) none of the mentioned
9. The Turing Church Thesis states that Turing Machines can solve any computable problem. (True/False) [BL-1]
10. The Language Accepted by Turing machine is called ____.
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Long Answer Questions

11. Construct a Turing Machine that can accept the string over $\{0,1\}$ containing even number of 1's. [BL-3]
12. Construct a Turing Machine to recognize the language: $\{ a^n b^n c^m \mid n, m \geq 1 \}$ [BL-3]
13. Design a TM to accept the language $L = \{ a^n b^n c \mid n > 1 \}$ [BL-3]
14. Design a TM to accept the set of all strings $\{0,1\}$ with 010 as substring. [BL-3]
15. Design a Turing Machine to accept the language $L = \{ 0^n 1^n \mid n \geq 1 \}$. [BL-3]

16. Explore the distinction between recursive and recursively enumerable languages. Define each type of language and discuss their properties, including decidability and recognizability. [BL-2]
17. Construct a Turing machine for $L = \{a^i b^j c^k \mid i * j = k, i, j, k \geq 1\}$. [BL-3]
18. Construct Turing machine for $L = \{a^n b^m a^{(n+m)} \mid n, m \geq 1\}$ [BL-3]
19. Explain the concept of a Turing Machine (TM) and how it serves as a fundamental model of computation. Describe its components, including the tape, read/write head, states, and transition function. [BL-2]
20. Construct TM for the addition function for the unary number system. [BL-3]
21. Implement a Turing machine that, given an input number **n** in binary representation, multiplies the number by 2. [BL-3]
22. Construct a TM for the language $L = \{(0^n 1^{2n}) \text{ where } n \geq 1\}$. [BL-3]
23. Construct a TM machine for checking the palindrome of the string of even length. [BL-3]
24. Construct a TM machine for checking the palindrome of the string of odd length. [BL-3]

MFT Unit 05

1. A multi track Turing machine can described as a 6-tuple (Q, X, S, d, q_0, F) where X represents:
 - a) input alphabet
 - b) tape alphabet
 - c) shift symbols
 - d) none of the mentioned
2. Which of the following is/are not true for recursively enumerable languages?
 - a) Partially decidable
 - b) Turing acceptable
 - c) Turing Recognizable
 - d) None of the mentioned
3. A Turing machine with several tapes is known as:
 - a) Multi-tape Turing machine
 - b) Poly-tape Turing Machine
 - c) Universal Turing machine
 - d) All of the mentioned
4. A multiple Turing machine is _____ powerful than a single tape Turing machine:
 - a) More
 - b) Less
 - c) None of the mentioned
5. Universal TM influenced the concept of:
 - a) Stored program computers
 - b) Interactive implementation of programming language

- c) Computability
- d) All of these

6. Which of the following are undecidable problems?

- a) Determining whether two grammars generate the same language
- b) Determining whether a grammar is ambiguous
- c) Both (a) and (b)
- d) None of the mentioned

7. Which one of the following is true for the given?

$A = \{(M, w) \mid M \text{ is a Turing machine that accepts string } w\}$

- a) A concrete undecidable problem

8. Which problem is undecidable:

- a) Finiteness problem for FSA's
- b) Membership problem for CFG's
- c) Equivalence problem for FSA's
- d) Ambiguity problem for CFG's

9. Which of the following statements are undecidable?[BL-2]

For a given Turing Machine M ,

- a) does M halt on an empty input tape
- b) does M halt for only inputs at all?
- c) is $L(M)$ regular? Context free? Turing decidable?
- d) all of the mentioned

10. Out of the three decision problems P_1 , P_2 , and P_3 , P_1 is decidable and P_2 is undecidable. The statement that holds true is:

- a) P_3 is decidable if P_1 is reducible to complement of P_2
- b) P_3 is decidable if P_1 is reducible to P_2
- c) P_3 is undecidable if P_1 is reducible to P_2
- d) P_3 is undecidable if P_2 is reducible to P_3

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11. Define a recursively enumerable language. Under what conditions can a recursively enumerable language be classified as recursive?
 12. Define a recursive language. How does it differ from a recursively enumerable language?
 13. Write short notes on the following topics:
 - i) Turing Reducibility
 - ii) Multi-Tape Turing Machine
 14. Write Short Note on NP-completeness. Explain the concept of NP-completeness with examples of well-known NP-complete problems.
 15. Explain the Halting Problem of a Turing Machine. Why is it undecidable? Provide a basic Explanation.
 16. Describe the model of a Non-Deterministic Turing Machine (NDTM). Explain how it differs from a deterministic Turing machine. Is an NDTM more powerful than a basic Turing Machine? Justify your answer.
 17. Explain the concept of a Multi-Tape Turing Machine. How does it function, and in what way does it differ from a single tape Turing Machine? Is it more powerful than a basic Turing Machine? Justify your answer.
 18. Explain the concept of a decidable problem. When do we say that a problem is decidable? Provide an example of an undecidable problem.
 19. Give an example of a language that is not recursively enumerable. Explain why this language does not fall under the category of recursively enumerable languages.
 20. Give an example of an undecidable problem. Explain why this problem is undecidable with appropriate reasoning.

21. Explain the concepts of recursive and recursively enumerable languages.
Provide a detailed comparison between these two types of languages and illustrate with suitable examples.
22. Prove that the union of two recursively enumerable languages is also recursively enumerable. Provide a formal proof.
23. Prove that the union of two recursive languages is also recursive. Provide a logical explanation with proof.
24. Prove that the following problem is undecidable: “Given two context-free grammars (CFGs) G_1 and G_2 , determine whether the intersection of their languages, $L(G_1) \cap L(G_2)$, is empty”.
25. Prove that the Post Correspondence Problem (PCP) is undecidable.
Formally define the PCP problem and provide a proof demonstrating its undecidability.

Unit 1 – MFT

1. Assume the R is a relation on a set A . aRb is partially ordered such that a and b are
 - a) Reflexive
 - b) Transitive
 - c) Symmetric
 - d) Reflexive and Transitive
2. A regular language over an alphabet Σ is one that cannot be obtained from the basic languages using the operation:
 - a) Union
 - b) Concatenation
 - c) Kleene*
 - d) All of the mentioned
3. The minimum number of states required to recognize an octal number divisible by 3 are/is:
 - a) 3
 - b) 4
 - c) 5
 - d) The number of elements in the set for the Language
4. $L = \{x \in \Sigma^* \mid \text{"length of } x \text{ is at most 2"}\}$ and $\Sigma = \{0,1\}$ is:
 - a) $\emptyset$
 - b) $\{\epsilon\}$
 - c) Finite
 - d) Infinite
5. Given $\Sigma = \{a, b\}$, $L = \{x \in \Sigma^* \mid x \text{ is a string combination}\}$. Σ^* represents which among the following?
 - a) $\{aa, ab, ba, bb\}$

- b) {aaa, aab, aba, abb}
- c) {aaa, aab, aba, abb, bbb}
- d) All of the mentioned

6. The Grammar can be defined as: $G = (V, \Sigma, P, S)$. In the given definition, what does S represent?

- a) Accepting State
- b) Starting Variable
- c) Sensitive Grammar
- d) None of these

7. State true or false:

Statement: $R \rightarrow RT$

$T \rightarrow a$ is an ambiguous grammar

- a) true
- b) can't say
- c) false
- d) none

8. What is the output for the given language?

Language: A set of strings over $\Sigma = \{a, b\}$ & takes as input and it prints 1 as an output "for every occurrence of a, b as its substring."

(INPUT: abaabb)

- a) 01010010
- b) 01010100
- c) 01101010
- d) 01010101

9. A language for which a DFA exists is a:

- a) Regular Language
- b) Non-Regular Language

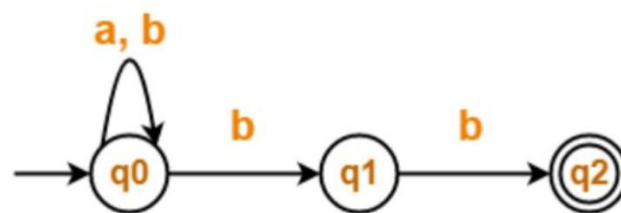
- c) May be Regular
- d) Cannot be said

10. When there are 2 finite states equivalent?

- a) Same number of transitions
- b) Same number of states
- c) Same number of states as well as transitions
- d) Both are final states

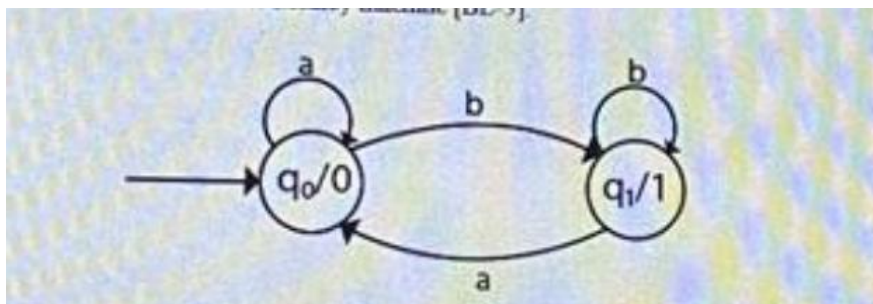
11. Explain transition diagram, transition table and transition function of DFA with suitable examples. [BL-2]

12. Define ϵ -transitions and convert the following NFA to DFA: [BL-3]

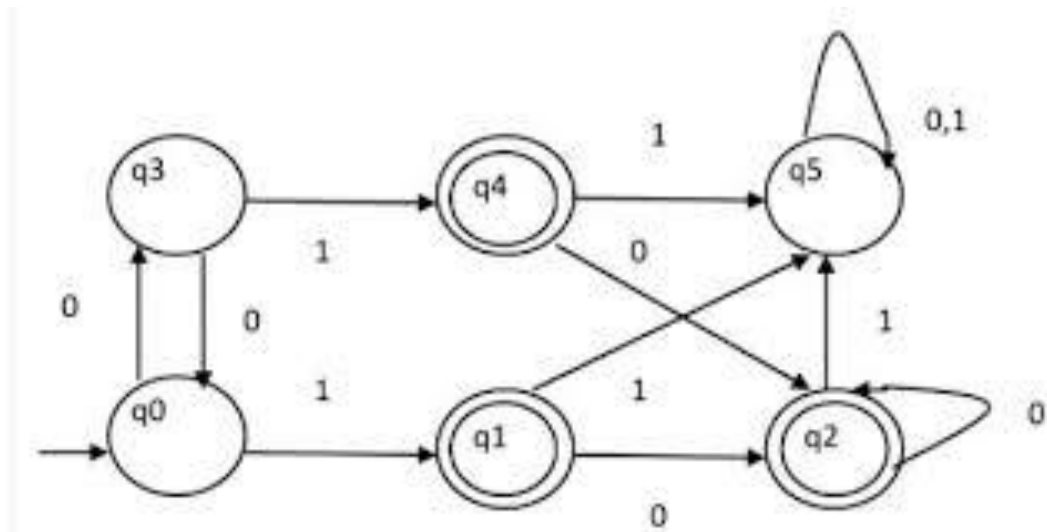


13. Define DFA, NFA, String, Language and Grammar with suitable explanation [BL-2]

14. Differentiate between Mealy and Moore machine also convert the following Moore machine into Mealy machine. [BL-3]



15. Minimize the below given DFA [BL-3]



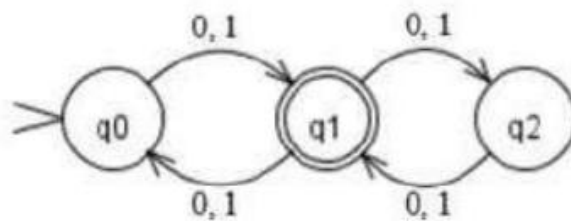
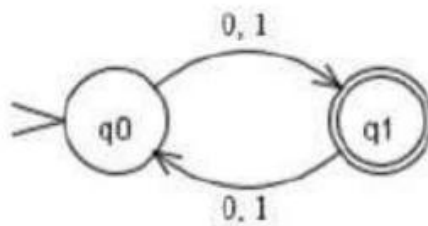
16. Obtain DFAs to accept strings of a's and b's having exactly one a.

[BL-3]

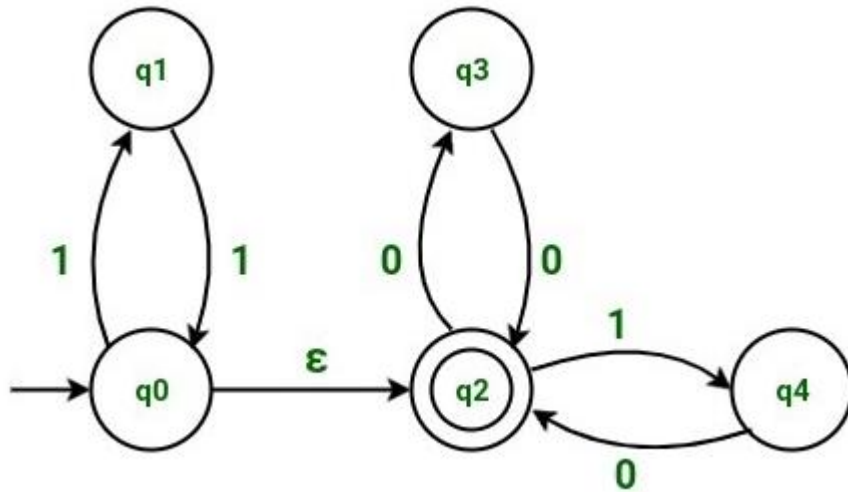
17. Draw a DFA for a string ending in '01' and 1 ending with 011. [BL-3]

18. Design a finite automata which performs addition of two binary numbers
also Draw a DFA to accept string of 0's and 1's ending with 011. [BL-3]

19. Proof that these two machines are equivalent or not? [BL-3]



20. Convert the following epsilon-NFA to NFA [BL-3]



21. Explain Myhill Nerode theorem with suitable example in detail [BL-3]